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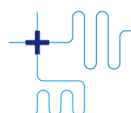
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Unexpected significant increase in bulk conductivity of a dielectric arising from charge injection

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Charge injection is a common phenomenon in heterostructures or devices containing metal-insulator interfaces under a voltage bias ranging from dielectric capacitors to electroluminescent and lasing devices. It is generally believed that charge injection only significantly increases the conductivity near the interfacial region or in capacitors with very thin dielectric layers. In this work, the impact of charge injection on bulk conductivity of a 0.5 mm thick Fe-doped SrTiO₃ single crystal is investigated with a combination of experimental impedance measurements and computational modelling. It is found that the interfacial charge injection may increase the predicted bulk conductivity of a dielectric by more than one order of magnitude as a consequence of Schottky barrier height lowering. *Published by AIP Publishing.* [<http://dx.doi.org/10.1063/1.4990677>]

Charge injection is of significant technological importance in the operation of semiconductor electronic devices.^{1–5} The ability to tune the amount of charges injected into a low conductivity bulk material by engineering the interfacial contacts enables a variety of applications.^{6–8} These range from light-emitting diodes, which demand the highest possible charge injection achieved by formation of Ohmic or tunnel contacts, to capacitors for which the lowest possible charge injection is achieved by formation of Schottky contacts.^{9,10}

A common approach to analyzing the presence of charge injection is to compare the current voltage behavior with existing theories such as Mott-Gurney law.^{1,3,5,11,12} A primary requirement for the validity of such an approach is that the sample is capable of reaching a steady state condition under a given applied voltage. However, there are dielectric materials such as doped dielectric oxides in which a steady state condition of the pristine state cannot be achieved due to resistance degradation^{13–15} arising from charged point defect migration.¹⁶ Whereas in existing resistance degradation models for these dielectric materials, charge injection was usually neglected and only local thermodynamic equilibrium among defects was assumed.¹³ This neglect of charge injection may induce some disagreements between the degradation behavior predicted from existing models^{13,14} and the experimental measurement. For instance, it has been previously shown that existing degradation models fail to predict the conductivity distribution immediately after the application of an electric field.¹⁷

In this work, a different approach is employed to identify the charge injection behavior, through the determination of the change in the spatial distribution of conductivity, i.e., the local conductivity using a combination of experimental measurements and computational modeling. Experimentally, the conductivity distribution is measured using *in-situ* impedance measurements since it can be used to separate the physical

processes characterized by a different relaxation frequency in the frequency domain.^{18–22} The Fe-doped single crystal SrTiO₃ is employed as a model system since it has been the most studied system for resistance degradation.^{13,14,17} Alternative techniques such as electron beam deflection²³ are also employed to make comparison to confirm the impedance measurements. Theoretically, the transport equation, the continuity equation, and the electrostatic equilibrium equation are incorporated^{1,2,4} and computationally solved to determine the space charge injection-induced conductivity distribution within the single crystal with mixed ionic and electronic charge carriers. Since the experimental procedure of impedance measurement is same as the previous work,^{17,18} hereafter we focus on the procedure of the modeling.

The current density j within the dielectric upon the application of an applied electric field E may be separated into two contributions^{2,4}

$$j = \left(\sum_i \sigma_i^0 + \sum_i \sigma_i^{\text{inj}}(x) \right) E(x) = \text{constant with } \sigma_i = z_i \mu_i c_i, \quad (1)$$

where σ is the conductivity within the dielectric and z_i , μ_i , and c_i are the charge, mobility, and concentration of the charge carrier species i . The superscript “inj” (injected) and “0” represent the conductivity arising from the injected charge carriers and that from the constant, thermal equilibrium values of the charge carriers in the bulk dielectric, respectively.²

For acceptor-doped mixed ionic and electronic dielectrics, the intrinsic and extrinsically doped defects satisfy the charge neutrality condition^{14,24–28}

$$e(p^0 + 2[V_O^{\bullet\bullet}] - n^0 - [A']) = 0, \quad (2)$$

where $[V_O^{\bullet\bullet}]$ is the oxygen vacancy concentration, p^0 the equilibrium hole concentration, n^0 the equilibrium electron concentration, and $[A']$ the acceptor concentration which is predominately iron in the case of Fe-doped SrTiO₃. Changes close to the interface due to contact formation are neglected

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since the extent of the bulk dielectric ($500\ \mu\text{m}$) is very large with respect to the Debye length ($<0.1\ \mu\text{m}$).^{1,29} This is also supported by the fact that the conductivity distribution of the pristine Fe-doped SrTiO_3 with the relevant thickness is homogeneous.^{17,18} For pristine Fe-doped SrTiO_3 , $\sum_i \sigma_i^0$ is dominated by the oxygen vacancy conductivity.³⁰

Assuming that the injected carriers are not compensated for by defects in the host material, a space charge region will be generated. The charge density distribution in this region can be calculated via the electrostatic equilibrium equation^{1,2,4}

$$\epsilon_0 \epsilon_r \frac{dE(x)}{dx} = e [p^{\text{inj}}(x) - n^{\text{inj}}(x)], \quad (3)$$

where ϵ_0 and ϵ_r represent the vacuum permittivity and relative permittivity of the dielectric, respectively, and e is the charge of an electron. The total electronic charge carrier concentrations are $n(x) = n^0 + n^{\text{inj}}(x)$ and $p(x) = p^0 + p^{\text{inj}}(x)$.

Due to the externally applied voltage, electrons can only be injected from the cathode/dielectric interface with lower potential, and holes can only be injected from the anode/dielectric interface with higher potential. The concentrations of

injected electrons and holes at their corresponding interfaces can be determined from their Schottky barrier heights,^{1,31} i.e.,

$$n|_{\text{Cathode}} = N_C \exp\left(-\frac{\phi_{B,n}}{k_B T}\right), \quad p|_{\text{Anode}} = N_V \exp\left(-\frac{\phi_{B,p}}{k_B T}\right), \quad (4)$$

with N_C and N_V the effective density of states in the conduction and valence bands, respectively, $\phi_{B,n}$ the Schottky barrier height for electrons at the cathode/dielectric interface, $\phi_{B,p}$ the Schottky barrier height for holes at the anode/dielectric interface, k_B the Boltzmann constant, and T the temperature.

The interaction between electrons and holes is described via the steady-state continuity equation^{1,32}

$$\frac{dj_n}{dx} + \frac{j_n}{j} \frac{dj_o}{dx} = eK [(n^{\text{inj}}(x) + n^0)(p^{\text{inj}}(x) + p^0) - n^0 p^0], \quad (5)$$

where K is the recombination rate constant.

Combining the above Eqs. (1)–(5), the dependence of the local electric field on the local electron concentration can be obtained as following:

$$\frac{dE(x)}{dn(x)} = \frac{e\mu_n E(x) [p(x) - p^0 - (n(x) - n^0)]}{\epsilon_0 \epsilon_r K (n(x)p(x) - n^0 p^0) - (\mu_n n(x) + 2j_n/j\mu_{V_o} [V_o^*]) e [p(x) - p^0 - (n(x) - n^0)]}, \quad (6)$$

which can be numerically solved, e.g., using the Runge-Kutta method. A computer code written in FORTRAN language for solving Eq. (6) with high efficiency is provided, which can be employed to solve the general electrostatic problems involving charge injection.

The proposed methodology is verified by comparing the numerical results with the analytic solution of a trap-free insulator with finite bulk conductivity.² However, the current voltage characteristics are not shown here since the main purpose of this work is to analyze the impact of injected charges on the local conductivity distribution under a fixed voltage.

In the modeling, most of the material parameters are from the experiment such as $T = 210^\circ\text{C}$, $V_{\text{DC}} = 40\ \text{V}$, $d = 500\ \mu\text{m}$, and $[\text{Fe}] = 5.58 \times 10^{18}\ \text{cm}^{-3}$. The oxygen vacancy concentration of $2.43 \times 10^{18}\ \text{cm}^{-3}$ is obtained from the defect chemistry calculation¹⁴ considering that the specimen was annealed at 900°C in an oxygen partial pressure of 2 Pa. At $T = 210^\circ\text{C}$, the relative permittivity from the measurement is $\epsilon_r = 220$, which agrees with previous experiment.¹⁷ The effective densities of states in the valence band and the conduction band are calculated via the effective mass of holes ($m_h^* = 12m_0$) and electrons ($m_e^* = 6m_0$) extracted from first-principles band structure calculations. Here, m_0 is the free electron mass. The remaining quantities are from the literature:²⁸ $\mu_n = 0.56\ \text{cm}^2\text{V}^{-1}\text{s}^{-1}$, $\mu_p = 0.41\ \text{cm}^2\text{V}^{-1}\text{s}^{-1}$, and $\mu_{V_o} = 2.20 \times 10^{-8}\ \text{cm}^2\text{V}^{-1}\text{s}^{-1}$. The recombination rate constant K is assumed to be $1.0 \times 10^{-8}\ \text{cm}^3\text{s}^{-1}$, and its effect on the local conductivity will be discussed below.

Figure 1(a) shows the experimental *in-situ* modulus data for the pristine state of a Fe-doped SrTiO_3 single crystal as

well as two stages shortly after the voltage is applied. The single modulus peak with relaxation frequency f_r shifts from 200 Hz to 2244 Hz after 130 s along with slight broadening.

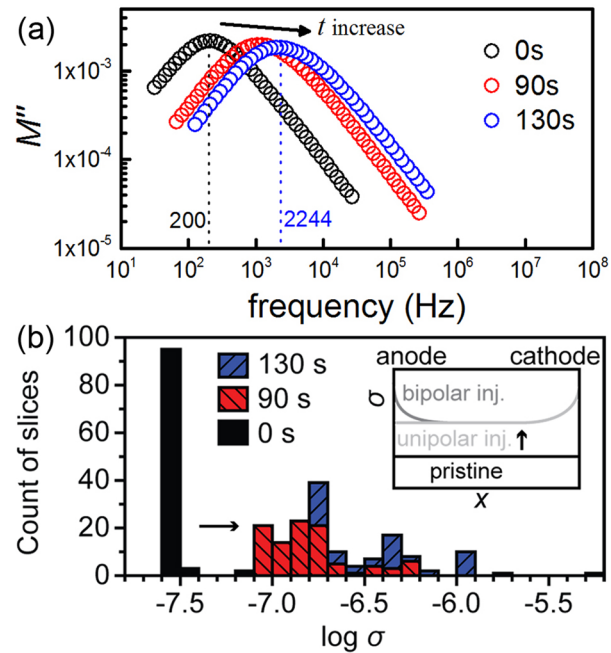


FIG. 1. (a) Experimental modulus data for the pristine state ($t=0\ \text{s}$) and states shortly after applying voltage ($t=90\ \text{s}$, $130\ \text{s}$). (b) Distributions of the conductivity fitted from the experimental measured modulus spectrum using the least squares method. The inset indicates the expected increase and alteration of the spatial distribution of conductivity from the pristine state to a condition governed by unipolar or bipolar charge injection.

Figure 1(b) displays the corresponding distribution of conductivity using a least squares fitting method¹⁸ with 100 equally spaced slices. It can be seen that the conductivity increases within the entire single crystal when a voltage is applied. The broadening of the modulus peak can be related to regions with slightly higher conductivity than others. While the conductivity values cannot be assigned to a specific region within the crystal from the modulus data, the high conductivity regions may be close to the electrode interfaces suggested by the work of Raith *et al.*²³ This distribution of conductivity is schematically illustrated in the inset of Fig. 1(b) assuming unipolar or bipolar charge injection. In the following, both cases will be discussed and compared with the experimental data.

Figure 2 displays the conductivity profiles $\sigma^{\text{inj}}(x)$ determined from the above model assuming that only electrons are injected via the cathode interface in a trap-free insulator with varied thickness, but fixed Schottky barrier height of $\phi_{B,n} = 0.5$ eV and a fixed ratio of applied voltage to thickness of 800 V/cm. For this unipolar charge injection, the conductivity at the cathode is fixed by the Schottky barrier height and exhibits the maximum value $\sigma_{\text{max}}^{\text{inj}}$. The number of injected electrons decays as a function of distance from the cathode so that a minimum value of $\sigma_{\text{min}}^{\text{inj}}$ is reached at the anode interface. The finite bulk conductivity σ^0 dominated by oxygen vacancies is given as a reference. By comparing the magnitude of $\sigma_{\text{min}}^{\text{inj}}$ and σ^0 , it is possible to determine if injected charges increase the total conductivity within the dielectric. The impact of injected charges is expected to decrease with increasing thickness of the dielectric. However, it is shown that with the sample conditions and properties given above, $\sigma_{\text{min}}^{\text{inj}} > \sigma^0$ is given even for a thickness of 500 μm . As a consequence, the amount of the injected charge carriers in the Fe-doped SrTiO₃ is sufficient to increase the conductivity throughout the entire sample.

Figure 3(a) shows the relaxation frequency of the impedance modulus calculated from unipolar charge injection with different Schottky barrier heights, compared with experimental measurements at 0 s, 90 s, and 130 s, respectively. The relaxation frequency determined from unipolar charge injection increases from 200 Hz to 1778 Hz with Schottky barrier height lowering. Figure 3(b) provides a summary of the impact of $\phi_{B,n}$ on $\sigma_{\text{min}}^{\text{inj}}$ and $\sigma_{\text{max}}^{\text{inj}}$. For the electron barriers higher than 0.92 eV, which is equivalent

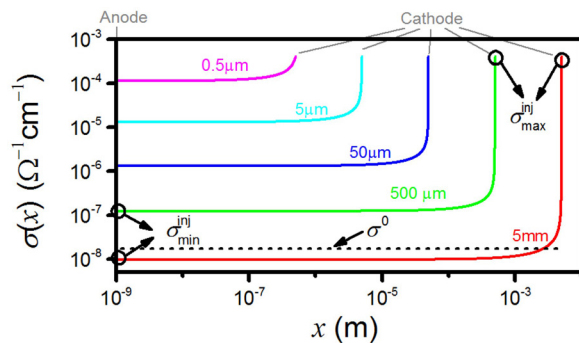


FIG. 2. Local conductivity induced by an unipolar charge injection of electrons in Fe-doped SrTiO₃ capacitors of varied thickness but the same Schottky barrier height of 0.5 eV and the same V^{app}/d . The black dashed line indicates σ^0 dominated by oxygen vacancies.

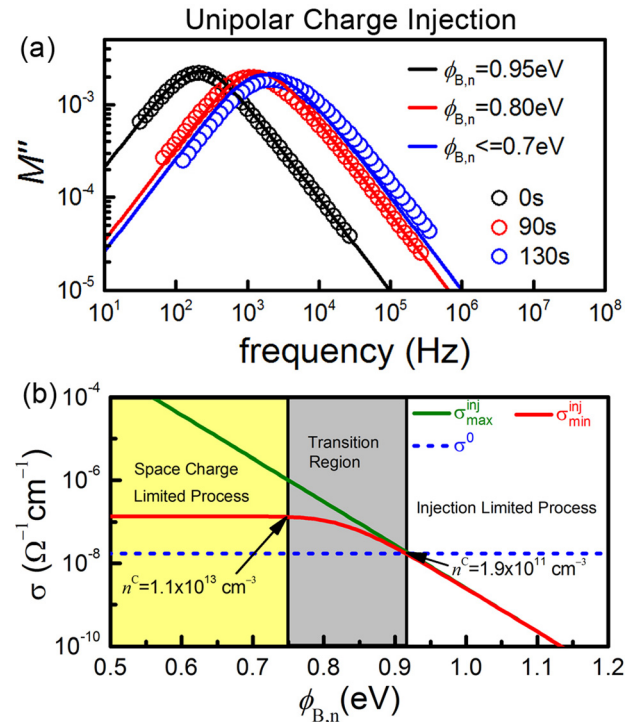


FIG. 3. (a) Comparison of modulus data between experiment (open circles) and simulation (solid lines) from the electron injection model. (b) Illustrations of the bulk conductivity as well as the calculated maximal and minimal local conductivity as a function of electron Schottky barrier height. In (b), n^C represents the electron concentration at the cathode/dielectric interface that marks the transition regime from an interface injection-limited ($\phi_B > 0.92$ eV) process to a space charge-limited ($\phi_B < 0.75$ eV) process.

to an electron density n^{inj} of lower than $1.9 \times 10^{11} \text{ cm}^{-3}$, the total conductivity is dominated by σ^0 . In the range of $0.92 \text{ eV} > \phi_{B,n} > 0.75 \text{ eV}$, both $\sigma_{\text{min}}^{\text{inj}}$ and $\sigma_{\text{max}}^{\text{inj}}$ increase with decreasing $\phi_{B,n}$. Below 0.75 eV, $\sigma_{\text{min}}^{\text{inj}}$ saturates and only $\sigma_{\text{max}}^{\text{inj}}$ increases further as the barrier decreases. For $\phi_{B,n} < 0.75 \text{ eV}$, the conductivity is dominated by a space charge limited process with a conductivity nine times higher than σ^0 , which can be calculated from the ratio of the relaxation frequency,¹⁸ i.e., $1778 \text{ Hz}/200 \text{ Hz} \cong 9$.

It is possible to assign the condition $\phi_{B,n} > 0.92 \text{ eV}$ to the pristine state at $t = 0$ s. This is reasonable since the platinum electrodes are known to form Schottky barriers of about 1.0 eV with SrTiO₃.^{33–36} It is also known that the Schottky barrier is decreased if the contact is annealed in a low oxygen partial pressure^{33,35,37} which is equivalent to an increase in oxygen vacancy concentration at the contact.^{14,17} Moreover, recent X-ray photoelectron spectroscopy (XPS) measurement indicated that the Schottky barrier height was decreased by 0.6 eV in Fe-doped SrTiO₃/RuO₂ interface during degradation.³⁸ Therefore, it is rational to expect that the electron barrier height should be decreased considering the fact that the oxygen vacancy concentration at the cathode interface can be increased slightly though slowly after the voltage is applied. With this in mind, the measured modulus data can be attributed to the decreasing $\phi_{B,n}$ values with increasing time.

Since the space charge injection increases the conductivity within the whole dielectric, the entire modulus peak shifts to higher frequencies. The result is in very good agreement with the experiment whereas previous modeling without

including the charge injection fails to reproduce the initial shift in the impedance modulus. It is important to point out that the range of $\phi_{B,n}$ used in the calculations lies within the possible range of barrier heights experimentally observed for the SrTiO₃/Pt interface.³³ Therefore, it can be concluded that the injected charge carriers may crucially impact the initial stage of the degradation process.

With only electrons injection considered, the maximum frequency for the modulus peak is 1778 Hz, which is slightly lower than 2244 Hz experimentally observed at 130 s. Now turn to the case where both electrons and holes are injected. Figure 4 shows the simulation results considering a bipolar charge injection with the same electron and hole barrier heights ($\phi_{B,n} = \phi_{B,p}$). As shown in Fig. 4(a), the impedance modulus results calculated from bipolar charge injection are similar to unipolar charge injection with respect to changes as a function of the Schottky barrier height. The relaxation frequency from bipolar charge injection with $\phi_{B,n} = \phi_{B,p} = 0.8$ eV is about 2454 Hz, compared with 1778 Hz and 2244 Hz from unipolar charge injection and experimental measurement at 130 s, respectively. While if the two barrier heights are further reduced to be smaller than 0.71 eV, the relaxation frequency can be up to 4200 Hz. Figure 4(b) shows the corresponding conductivity profiles. It can be seen that the conductivity is increased at the vicinity of both electrode contacts, which qualitatively matches the conductivity profile observed by Raith *et al.*²³ Figure 4(c) shows $\sigma_{\min}^{\text{inj}}$ and $\sigma_{\max}^{\text{inj}}$ as function of the Schottky barrier heights. It can be seen that the transition range for the Schottky barrier heights between the injection-limited

(above 0.95 eV) and space charge-limited (below 0.71 eV) transport is increased. Thus, a lower barrier height is needed to reproduce the measured modulus feature.

Figure 4(d) presents the individual contributions of electron, hole, and oxygen vacancy conductivity that impact the total conductivity with barrier heights of 0.6 eV for both the anode and cathode. The minimum conductivity $\sigma_{\min}^{\text{inj}}$ with bipolar injection is now inside the dielectric. There is also a slight asymmetry of the conductivity evolution due to the fact that $N_V > N_C$ and $\mu_n > \mu_p$. Figure 4(d) illustrates the effect of the recombination rate constant K which is unknown for the indirect-bandgap semiconductor SrTiO₃. For simplicity, Eq. (5) assumes a direct band-gap recombination and generation process in which the recombination is accomplished directly by the transition of an electron from the conduction band to the valence band accompanied by a radiative process or Auger process. In contrast, for an indirect band-gap, the recombination has to be accomplished by at least two processes depending on the trapping levels. Parameters involved in the indirect recombination process such as trap levels, concentration of traps, net recombination rates, capture coefficients for electrons and holes^{39–41} are rarely available experimentally. Therefore, in this work, we use the direct band-to-band transition to approximate the recombination process in the SrTiO₃ single crystal although it has an indirect band-gap. Figure 4(d) demonstrates that while the local electron and hole conductivity strongly depend on the value of K , the total conductivity is weakly dependent on K . Since we are mainly concerned with the total conductivity,

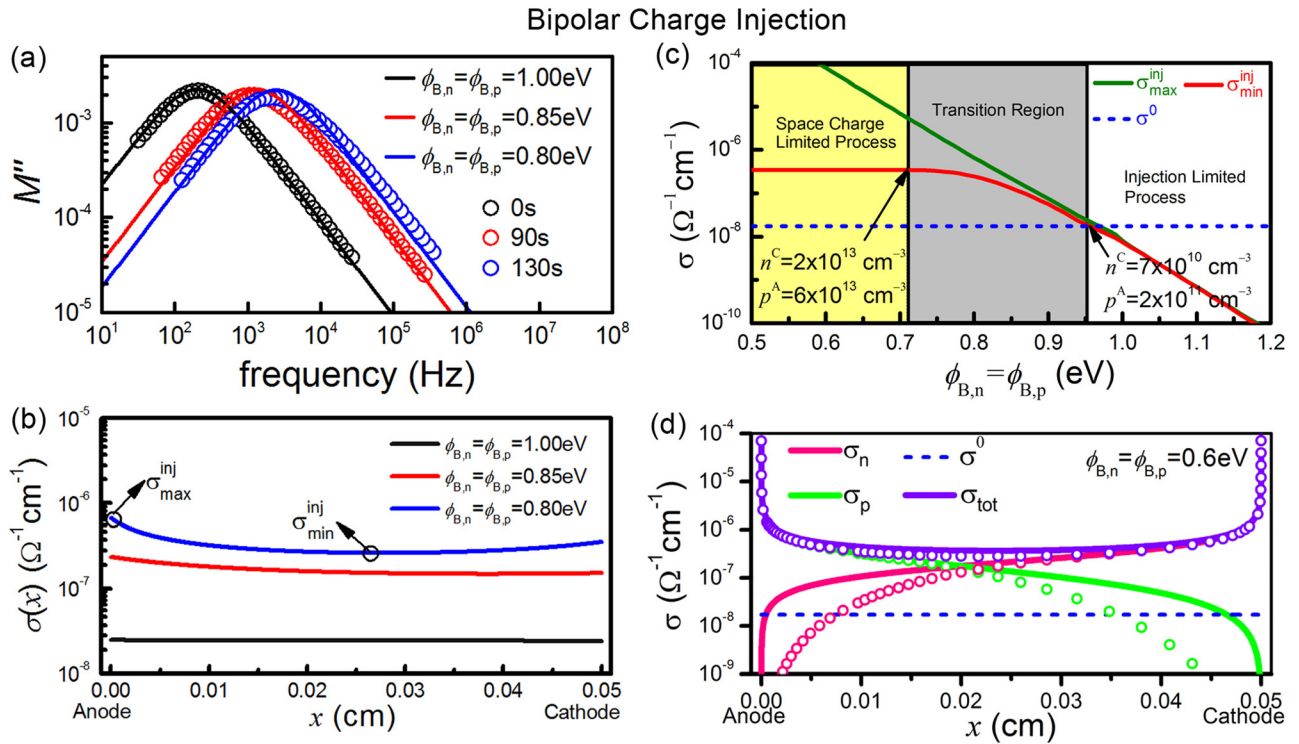


FIG. 4. (a) Comparison of modulus data between experiment (open circles) and simulation (solid lines) from bipolar charge injection model with equal barrier heights. (b) Corresponding conductivity profiles related to the modulus data shown in (a). (c) Illustration of σ^0 , $\sigma_{\min}^{\text{inj}}$, and $\sigma_{\max}^{\text{inj}}$ as a function of Schottky barrier height. In (c) n^C and p^A represent the required concentration of injected electrons and holes at each interface that mark the transition regime from an interface injection-limited ($\phi_B > 0.95$ eV) process to a space charge-limited ($\phi_B < 0.71$ eV) process. (d) Spatial distribution of individual conductivity from different charge carriers for the case of equal electron and hole barrier heights of 0.6 eV. Conductivities are calculated with a recombination rate constant K of 1.0×10^{-8} (lines) and $2.5 \times 10^{-8} \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$ (symbols), respectively.

the conclusions that we derived in this work are rather independent of the choice for the value of K .

With the bipolar charge injection, the conductivity in the dielectric may be increased up to 21 times with respect to σ^0 , which can be calculated from the ratio of the relaxation frequency,¹⁸ i.e., $4200 \text{ Hz}/200 \text{ Hz} = 21$. The modulus data as well as the distribution of local conductivity agree well with experimental measurements. This strongly implies that space charge injection is important and should be considered when modeling the dielectric degradation process.

In summary, we investigated the impact of space charge injection on the early stage of dielectric degradation by experimentally measuring and computationally modeling the local conductivity distribution within a model dielectric, the Fe-doped SrTiO₃ single crystal. It is found that the space charge injection plays an unexpectedly significant role in the bulk conductivity increase even when the thickness of the dielectric capacitor is as thick as $500 \mu\text{m}$, leading to an order of magnitude increase in conductivity compared to that without charge injection. The present work not only provides a general methodology for understanding the physics of charge injection but also suggests that even in bulk systems the charge injection may play an important role. Therefore, it is important to take charge injection into account when studying the degradation of both bulk and thin-film dielectrics. The attached computer code based on the proposed methodology can be employed to solve the bipolar and unipolar charge injection equations for any direct-bandgap semiconductors, insulators, and diodes with mixed electronic and ionic conductivities.⁴²

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¹S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices* (John Wiley & Sons, Inc., Hoboken, NJ, 1985).

²M. A. Lampert and P. Mark, *Current Injection in Solids* (Academic Press, New York, 1970).

³K. C. Kao, *Dielectric Phenomena in Solids: With Emphasis on Physical Concepts of Electronic Processes* (Academic Press, 2004).

⁴R. H. Tredgold, *Space Charge Conduction in Solids* (Elsevier Publishing Company, 1966).

⁵G. W. Dietz and R. Waser, *Thin Solid Films* **299**, 53 (1997).

- ⁶F. Guo, B. Yang, Y. Yuan, Z. Xiao, Q. Dong, Y. Bi, and J. Huang, *Nat. Nanotechnol.* **7**, 798 (2012).
- ⁷B. Mashford, M. Stevenson, Z. Popovic, C. Hamilton, Z. Zhou, C. Breen, J. Steckel, V. Bulovic, M. Bawendi, S. Coe-Sullivan, and P. T. Kazlas, *Nat. Photonics* **7**, 407 (2013).
- ⁸R. J. Walters, G. I. Bourianoff, and H. A. Atwater, *Nat. Mater.* **4**, 143 (2005).
- ⁹O. Auciello, *J. Appl. Phys.* **100**, 051614 (2006).
- ¹⁰J. F. Scott, *Science* **315**, 954 (2007).
- ¹¹Y. Shen, M. W. Klein, D. B. Jacobs, J. Campbell Scott, and G. G. Malliaras, *Phys. Rev. Lett.* **86**, 3867 (2001).
- ¹²W. R. Silveira and J. A. Marohn, *Phys. Rev. Lett.* **93**, 116104 (2004).
- ¹³T. Baiatu, R. Waser, and K.-H. Härdtl, *J. Am. Ceram. Soc.* **73**, 1663 (1990).
- ¹⁴J.-J. Wang, H.-B. Huang, T. J. M. Bayer, A. Moballegh, Y. Cao, A. Klein, E. C. Dickey, D. L. Irving, C. A. Randall, and L.-Q. Chen, *Acta Mater.* **108**, 229 (2016).
- ¹⁵R. Waser, T. Baiatu, and K.-H. Härdtl, *J. Am. Ceram. Soc.* **73**, 1654 (1990).
- ¹⁶R. Waser and M. Klee, *Integr. Ferroelectr.* **2**, 23 (1992).
- ¹⁷T. J. M. Bayer, J.-J. Wang, J. J. Carter, A. Moballegh, J. Baker, D. L. Irving, E. C. Dickey, L.-Q. Chen, and C. A. Randall, *Acta Mater.* **117**, 252 (2016).
- ¹⁸T. J. M. Bayer, J. Wang, J. J. Carter, L.-Q. Chen, and C. A. Randall, in 2016 IEEE International Symposium on Applications of Ferroelectrics and European Conference on Application of Polar Dielectrics (2016).
- ¹⁹E. Barsouk and J. R. Macdonald, *Impedance Spectroscopy Theory, Experiment and Applications*, 2nd ed. (John Wiley & Sons, Inc., Hoboken, NJ, 2005).
- ²⁰J. T. S. Irvine, D. C. Sinclair, and A. R. West, *Adv. Mater.* **2**, 132 (1990).
- ²¹R. Waser and R. Hagenbeck, *Acta Mater.* **48**, 797 (2000).
- ²²L. G. Escrig, M. Prades, H. Beltran, E. Cordoncillo, N. Maso, and A. R. West, *J. Am. Ceram. Soc.* **97**, 2815 (2014).
- ²³A. Raith, P. Reijnen, and R. Waser, *Phys. Chem. Chem. Phys.* **92**, 1516 (1988).
- ²⁴R. Moos, W. Menesklou, and K. H. Härdtl, *Appl. Phys. A* **61**, 389 (1995).
- ²⁵N.-H. Chan, R. K. Sharma, and D. M. Smyth, *J. Electrochem. Soc.* **128**, 1762 (1981).
- ²⁶R. M. Waser, *J. Am. Ceram. Soc.* **72**, 2234 (1989).
- ²⁷G. M. Choi, H. L. Tuller, and D. Goldschmidt, *Phys. Rev. B* **34**, 6972 (1986).
- ²⁸I. Denk, W. Münch, and J. Maier, *J. Am. Ceram. Soc.* **78**, 3265 (1995).
- ²⁹R. A. De Souza, F. Gunkel, S. Hoffmann-Eifert, and R. Dittmann, *Phys. Rev. B* **89**, 241401(R) (2014).
- ³⁰R. A. Maier and C. A. Randall, *J. Am. Ceram. Soc.* **99**, 3350 (2016).
- ³¹P. A. Torpey, *J. Appl. Phys.* **56**, 2284 (1984).
- ³²W. Shockley, *Electrons and Holes in Semiconductors* (D. Van Nostrand Company, Inc., Princeton, NJ, 1950).
- ³³A. Klein, *J. Am. Ceram. Soc.* **99**, 369 (2016).
- ³⁴S. Li, F. Chen, R. Schafrank, T. J. M. Bayer, K. Rachut, A. Fuchs, S. Siol, M. Weidner, M. Hohmann, V. Pfeifer, J. Morasch, C. Ghinea, E. Arveux, R. Günzler, J. Gassmann, C. Körber, Y. Gassenbauer, F. Säuberlich, G. V. Rao, S. Payan, M. Maglione, C. Chirila, L. Pintilie, L. Jia, K. Ellmer, M. Naderer, K. Reichmann, U. Böttger, S. Schmelzer, R. C. Frunza, H. Uršič, B. Malič, W. Bin Wu, P. Erhart, and A. Klein, *Phys. Status Solidi RRL* **8**, 571 (2014).
- ³⁵R. Schafrank, S. Payan, M. Maglione, and A. Klein, *Phys. Rev. B* **77**, 195310 (2008).
- ³⁶N. Ohashi, H. Yoshikawa, Y. Yamashita, S. Ueda, J. Li, H. Okushi, K. Kobayashi, and H. Haneda, *Appl. Phys. Lett.* **101**, 251911 (2012).
- ³⁷T. Kawada, N. Iizawa, M. Tomida, A. Kaimai, K. Kawamura, Y. Nigara, and J. Mizusaki, *J. Eur. Ceram. Soc.* **19**, 687 (1999).
- ³⁸R. Giesecke, R. Hertwig, T. J. M. Bayer, C. A. Randall, and A. Klein, "Modification of the Schottky barrier height at the RuO₂ cathode during resistance degradation of Fe-doped SrTiO₃," *J. Am. Ceram. Soc.* published online.
- ³⁹C. Sah, R. Noyce, and W. Shockley, *Proc. IRE* **45**, 1228 (1957).
- ⁴⁰R. N. Hall, *Phys. Rev.* **87**, 387 (1952).
- ⁴¹W. Shockley and W. T. Read, *Phys. Rev.* **87**, 835 (1952).
- ⁴²See <http://www.ems.psu.edu/~chen/package/download.html> for the computational code for charge injection can be downloaded from the link.